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| **Fiscal Year:** | FY 2020 | **Task Last Updated:** | FY 06/17/2020 |
| **PI Name:** | Dannemiller, Karen Ph.D. | | |
| **Project Title:** | Humidity and Microbial Growth in ISS Dust (HUMID) | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
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| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | (1) Translational (Countermeasure) Potential<br>(2) Bioregenerative Life Support | | |
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| **Comments:** | | | |
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| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA ARC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Horack, John Ph.D. ( Ohio State University )<br>Meyer, Marit Ph.D. ( NASA Glenn Research Center ) | | |
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| **Task Description:** | In space, astronauts are confined in spacecraft with recirculated air and altered microbial exposures. These exposures have important implications for astronaut health (allergens, pathogens) [1], spacecraft integrity (prevention of microbial-induced corrosion such as on Mir [2]), and planetary protection from microbes. Microbial communities on the International Space State (ISS) have been characterized [3], [4], and this project would expand upon that work with novel chamber experiments to allow prediction of microbial communities and functions under different conditions. Moisture is a critical factor in determining the presence and growth of microorganisms. In fact, our recent work demonstrated that microbial growth in house dust can make a substantial contribution to human exposure under elevated relative humidity conditions (>80% RH) [5]. Moisture is inherent to an occupied spacecraft environment because water is necessary for life. Astronauts sweat, bathe, and exhale moisture, and the resulting water vapor and condensation occurs on the ISS and will occur on all future crewed missions. In fact, Mir was damaged by fungi that could degrade metals, glass, or plastics prior to its decommissioning [2]. Microbial growth may occur directly on surfaces, but also likely occurs in the dust. The dust represents a rich source of organic material and water is most likely the main factor limiting growth. Furthermore, dust is an important source of human exposure and could potentially contaminate other planets.<br><br>However, microbial growth on spacecraft is only beginning to be characterized [6], and we do not know which microbes will grow in the dust itself. Areas of high moisture could potentially harbor allergens, pathogens, and/or microbes capable of material degradation and planetary contamination. Currently, we have characterized neither a healthy nor unhealthy spacecraft microbiome.<br><br>This project will utilize vacuumed dust to be returned from the ISS in 2018 from collection in Divert Unwanted Space Trash (DUST). The goal of this project is to characterize microbial communities and function that may occur on the ISS in areas of high relative humidity and condensation.<br><br>Aim 1: Characterize microbial growth in ISS dust under different relative humidity conditions in a laboratory chamber study. Dust from the ISS will be incubated under varying relative humidity conditions [5]. We will characterize microbial growth using DNA-based methods to quantify growth and determine microbial communities present in both the original dust and incubated samples.<br><br>Aim 2: Identify potential microbial function that can occur in dust onboard the ISS. We will extract RNA from a subset of samples for metatranscriptomic analysis of fungal communities. Our yet unpublished work demonstrates that microbial function occurs even at 50% RH in house dust.<br><br>Our results will go beyond microbial measurement to predict moisture-related changes to the indoor microbiome and astronaut exposure. Our work will provide added value to microbial data collected previously on the ISS. For instance, Microbial Tracking 1 ( https:// ) characterized airborne and surface-associated populations of microorganisms aboard the ISS. HUMID complements this data through use of a chamber study observing fundamental microbial processes that may occur under conditions of increased moisture in occupied spacecraft. We need to understand how to characterize the spacecraft microbiome as either healthy or unhealthy. Ultimately we will be able to develop improved microbial clearance protocols that are more relevant to astronaut health, planetary protection, and spacecraft integrity. Our design decisions in space have important implications in controlling the microbes that are present. By understanding the fundamental processes that occur in this environment, we can design for a healthy microbiome instead of being subject to unintended consequences.<br><br>References<br><br>[1] M. J. Mendell, A. G. Mirer, K. Cheung, M. Tong, and J. Douwes, “Respiratory and Allergic Health Effects of Dampness, Mold, and Dampness-Related Agents: A Review of the Epidemiologic Evidence,” Environ. Health Perspect., vol. 119, no. 6, pp. 748–756, 2011.<br><br>[2] N. Novikova, P. De Boever, S. Poddubko, E. Deshevaya, N. Polikarpov, N. Rakova, I. Coninx, and M. Mergeay, “Survey of environmental biocontamination on board the International Space Station,” Res. Microbiol., vol. 157, no. 1, pp. 5–12, Jan. 2006.<br><br>[3] N. K. Singh, A. Blachowicz, A. Checinska, C. Wang, and K. Venkateswaran, “Draft Genome Sequences of Two Aspergillus fumigatus Strains, Isolated from the International Space Station.,” Genome Announc., vol. 4, no. 4, Jul. 2016.<br><br>[4] A. Checinska, A. J.Probst, P. Vaishampayan, J. R. White, D. Kumar, V. G. Stepanov, G. E. Fox, H. R. Nilsson, D. L. Pierson, J. Perry, and K. Venkateswaran, “Microbiomes of the dust particles collected from the International Space Station and Spacecraft Assembly Facilities,”Microbiome, vol. 3, no. 1, p. 50, Dec. 2015.<br><br>[5] K. C. Dannemiller, C. J. Weschler, and J. Peccia, “Fungal and bacterial growth in floor dust at elevated relative humidity levels,” Indoor Air, no. May, pp. 1–10, 2016.<br><br>[6] C. M. Ott, R. J. Bruce, and D. L.Pierson, “Microbial Characterization of Free Floating Condensate Aboard the Mir Space Station,” Microb. Ecol., vol. 47, no. 2, pp. 133–136, 2004. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This work broadens understanding of microbial growth in dust, which is relevant both on the International Space Station (ISS) and in many environments on Earth. This includes homes, offices, schools, other buildings, and specialized environments such as aircraft and submarines. Our work provides a framework with which to model this growth and allow potential prediction of conditions that could lead to adverse outcomes. The growth in dust has implications both for material corrosion and human health. |

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| | The goal of this project is to characterize microbial communities and microbial function that could possibly occur on the International Space Station (ISS) in areas of high relative humidity (RH) and condensation. We will achieve this goal through the following aims:<br>Aim 1: Characterize microbial growth in ISS dust under different relative humidity conditions in a laboratory chamber study. Dust from the ISS has been returned to Earth following an existing collection experiment called “Divert Unwanted Space Trash (DUST).” This dust will be incubated under varying relative humidity conditions (50%, 60%, 70%, 80%, 85%, 90%, and 100% RH) using established protocols. We will characterize microbial growth using quantitative Polymerase Chain Reaction (qPCR) and high-throughput DNA sequencing to quantify growth and determine microbial communities present in both the original dust and incubated samples.<br><br>Aim 2: Identify potential microbial function that can occur in dust onboard the ISS. We will incubate a subset of samples at 50%, 85%, and 100% RH and extract RNA for metatranscriptomic analysis of fungal communities. Our previous work demonstrates that microbial function occurs even at 50% RH in house dust. |
| **Task Progress:** | At this stage in the project in November 2019, we have received dust from two different vacuum bags that were returned from the ISS. The focus of our work has been on performing aim 1 experiments. Dust incubations for both ISS vacuum bags will be completed by mid-December 2019. Some of these incubations have been completed and total fungal growth has been quantified. Visual analyses of microbial growth in the ISS dust has also been performed via scanning electron microscopy (SEM). The first vacuum bag samples have been prepared for DNA sequencing. Our goal is to have all the sample incubations completed before the end of 2019 and DNA sequencing will take place at the beginning of 2020.<br><br>We have developed our pipeline for aim 2 and have determined the correct amount of dust needed in the extraction process to produce enough quality RNA for microbial function analysis. The first set of samples in triplicate have been incubated at 50%, 85%, and 100% RH and RNA has been extracted. Quality control analysis using bioanalyzer trace results has revealed that our extractions have produced enough RNA for metatranscriptome sequencing. We have also been afforded enough dust to allow for incubations for a small subset of samples to undergo a chemical analysis. We plan to target chemicals such as flame retardants such as organophosphate compounds and brominated flame retardants. |
| **Bibliography Type:** | Description: (Last Updated: 12/08/2025) |
| **Papers from Meeting Proceedings** | Bope A, Haines SR, Nastasi N, Horack JM, Meyer ME, Dannemiller KC. "Moisture availability and microbial activation in spacecraft." 70th International Astronautical Congress (IAC), Washington, DC, October 21-25, 2019.<br>70th International Astronautical Congress (IAC), Washington, DC, October 21-25, 2019. Paper IAC-19.A1.8.3. , Oct-2019 |
| **Papers from Meeting Proceedings** | Haines SR, Bope A, Nastasi N, Horack JM, Meyer ME, Dannemiller KC. "Measurement of fungi and bacteria from dust collected on the International Space Station (ISS)." 49th International Conference on Environmental Systems, Boston, MA, July 7-11, 2019.<br>49th International Conference on Environmental Systems, Boston, MA, July 7-11, 2019. ICES paper ICES-2019-102. , Jul-2019 |